

- 6 (a) (i) Identify the precise location in a cholinergic synapse that is described in each of the statements.

A region that contains many mitochondria.

.....

A region where exocytosis of ACh occurs.

.....

A region that contains voltage-gated channel proteins.

.....

A region that contains ligand-gated channel proteins.

.....

[4]

- (ii) Outline the roles of synapses in the nervous system.

.....

.....

.....

..... [2]

- (b) Some insecticides contain compounds called organophosphates. Organophosphates can act as enzyme inhibitors in neuromuscular junctions.

People who use organophosphates and do not follow safety guidelines are at risk of organophosphate poisoning, which affects muscle function.

Suggest how organophosphates could affect muscle function in humans.

.....

.....

.....

.....

.....

.....

.....

..... [4]

(c) Outline the role of the sarcoplasmic reticulum in the contraction of striated muscle.

.....

.....

.....

.....

..... [2]

[Total: 12]